

Report

This project analyzes e-commerce product data to help the business owner evaluate product performance, inventory health, and pricing effectiveness. The goal is to identify strong and weak brands and categories, spot products that are out of stock or underperforming, and recommend actions that can improve overall business performance.

1. Top 10 Products by Popularity Score

Identifying the platform's highest-engagement items to optimize marketing exposure and inventory readiness.

	name character varying (100) 	brand character varying (50) 	popular numeric (10,2) 
1	Sony Headphones	Sony	24745.00
2	iPhone 13	Apple	24416.70
3	Dell Laptop	Dell	24305.00
4	OnePlus 11	OnePlus	24304.00
5	Nike Air Max	Nike	24160.00
6	Adidas Ultraboost	Adidas	24015.00
7	Levi's Jeans	Levi's	23965.00
8	Wildcraft Backpack	Wildcraft	23947.20
9	Puma Sneakers	Puma	23918.40
10	OnePlus 11	OnePlus	23902.20

- **Electronics & Apparel Dominance:** Premium tech (Sony, Apple, Dell, OnePlus) and major footwear/apparel brands dominate customer interest.
- **Severe Supply Risk:** 8 out of the top 10 most popular products are either completely *Out of Stock* or on *Limited Stock*, signalling major potential revenue loss.

Action recommendation:

- **Urgent Inventory Replenishment:** Prioritize restocking Wildcraft, OnePlus, Sony, and Apple items immediately to capture unmet demand.

- **Pause Ad Spend:** Temporarily reduce marketing spend on the limited/out-of-stock items to prevent wasting ad budget on products customers cannot buy.

2. Top 10 products by review count.

Identifying products with the highest customer engagement and post-purchase feedback volume to determine market saturation and brand loyalty.

	product character varying (100) 🔒	review integer 🔒
1	iPhone 13	5000
2	Nike Air Max	4997
3	H&M Sweatshirt	4995
4	Wildcraft Backpack	4989
5	H&M Sweatshirt	4987
6	iPhone 13	4983
7	OnePlus 11	4983
8	Puma Sneakers	4983
9	Nike T-shirt	4978
10	iPhone 13	4971

- **High Engagement/Mature Products:** Products like the *iPhone 13* and *H&M Sweatshirt* appear multiple times, suggesting these are high-velocity items with a large, active user base.
- **Feedback Density:** The top 10 products are tightly clustered (all between 4,971 and 5,000 reviews), indicating a consistent level of high consumer interaction across these core product lines.

Action recommendation:

- **Sentiment Analysis:** Perform a sentiment analysis on these high-review products to identify common customer pain points or praise, which can inform future product iterations.
- **Leverage Social Proof:** Use these high-review counts in marketing materials (e.g., "Join 5,000+ satisfied customers") to drive conversion for newer, lower-reviewed items.

3. Top 10 Products by Rating with Minimum Review Threshold

Identifying flawless, highly-rated products while filtering out single-buyer 5.0 ratings to find true customer favorites with verified social proof.

	name character varying (100) 🔒	brand character varying (50) 🔒	total_people_review integer 🔒	rating numeric (3,1) 🔒
1	Wildcraft Backpack	Wildcraft	3080	5.0
2	RayBan Sunglasses	RayBan	2489	5.0
3	Wildcraft Backpack	Wildcraft	3447	5.0
4	Reebok Shoes	Reebok	3406	5.0
5	Nike Air Max	Nike	2640	5.0
6	H&M Sweatshirt	H&M	3766	5.0
7	Wildcraft Backpack	Wildcraft	2541	5.0
8	Adidas Hoodie	Adidas	1831	5.0
9	Reebok Shoes	Reebok	2693	5.0
10	Wildcraft Backpack	Wildcraft	914	5.0

- **Elite Catalysts:** These are your top-tier performing products; achieving a perfect 5.0 rating with thousands of reviews indicates extreme product-market fit.
- **Wildcraft Outperformance:** Wildcraft backpacks dominate this elite category, but half of their top-rated variants are currently unpurchasable.

Action recommendation:

- **Gold Standard Protection:** Guard the stock of these exact items aggressively; they are conversion engines that guarantee high customer satisfaction.
- **"Best Seller" Badging:** Tag these items with exclusive web badges (e.g., "Top Rated") to capitalize on their massive, unblemished customer approval scores.

4: Brand-wise average rating, reviews, popularity, and stock distribution.

Evaluating aggregate brand performance and inventory stability across the entire platform to identify top-performing partners and chronic supply chain offenders.

	brand character varying (50)	average_rating numeric	review numeric	popularity numeric	count_in_stock bigint	count_limited_stock bigint	count_out_of_stock bigint
1	Dell	4.21	2567.20	10748.30	22	29	15
2	Apple	4.19	2619.08	11119.53	28	25	30
3	OnePlus	4.17	2591.16	10852.07	27	19	31
4	Wildcraft	4.16	2645.79	11044.13	53	40	53
5	H&M	4.16	2561.60	10735.30	28	29	25
6	American Tourister	4.15	2565.98	10680.59	41	48	40
7	RayBan	4.15	2404.90	9967.97	60	34	43
8	Sony	4.13	2469.40	10207.61	25	29	19
9	Levi's	4.10	2747.40	11191.69	24	31	22
10	Adidas	4.10	2398.08	9884.49	61	59	83
11	Bost	4.08	2367.61	9644.56	20	25	25
12	Nike	4.08	2545.72	10430.47	66	68	57
13	Zara	4.08	2899.10	11626.62	23	22	33
14	Samsung	4.07	2467.30	10114.09	22	25	27
15	Fossil	4.05	2327.59	9546.70	54	34	40
16	HP	4.04	2633.89	10623.21	24	20	20
17	Reebok	4.04	2559.18	10361.42	36	40	40
18	Puma	4.01	2398.70	9637.04	69	78	59

- **Quality Leaders vs. High Demand:** *Dell* holds the highest customer satisfaction (4.21), while *Zara* and *Levi's* lead the platform in consumer demand, commanding the highest average engagement and popularity scores.

- **The Stockout Red Zone:** *Adidas* (83 items out of stock) and *Wildcraft* (53 items out of stock) are experiencing severe operational bottlenecks, choking off high-popularity catalogs. Tech leaders *Apple* and *OnePlus* also have more items entirely out of stock than fully available.

Action recommendation:

- **Supplier SLA Enforcement:** Impose stricter fulfillment SLA metrics on *Adidas*, *Wildcraft*, and *Apple* to lower their high share of out-of-stock items.

- **Capitalize on Fashion Traffic:** Increase promotional focus on *Zara* and *Levi's* as they naturally generate the highest user traction and interaction metrics.

5: Category-wise average price, discount, and popularity.

Assessing macro-level performance across main product categories to evaluate pricing structures, discount distributions, and overall customer engagement.

	category character varying (50)	average_price numeric	average_discount_percentage numeric	average_discount_price numeric	average_popularity numeric
1	Electronics	45574.40	39.71	27358.47	10486.23
2	Footwear	5475.10	37.91	3414.13	9949.29
3	Accessories	3777.81	37.55	2337.33	10329.31
4	Fashion	2793.90	39.19	1711.71	10712.11

- **High-Volume vs. High-Value:** *Fashion* is the most popular category on average (10,712.11) despite having the lowest average transaction value, showing high customer interest and purchase frequency.
- **Aggressive Electronics Discounting:** *Electronics* carries the highest average price and the steepest discount percentage (~39.71%), indicating that tech sales are highly dependent on heavy promotional slashes to drive engagement.

Action recommendation:

- **Optimize Fashion Margins:** Since *Fashion* naturally pulls the highest popularity, consider slightly reducing the 39.19% average discount on trending items to recover profit margins without losing traffic.
- **Bundle Accessories with Electronics:** Create cross-category bundle promotions (e.g., matching a high-value electronic item with highly popular accessories) to increase the average order value (AOV).

6: Subcategory-wise product performance ranking.

Breaking down macro categories into specific subcategories to understand exactly which product types are driving the highest popularity, interaction volume, and customer satisfaction.

	performer_rank bigint	subcategory character varying (50)	popular numeric	total_review bigint	average_rating numeric
1	1	Shirt	11626.62	226130	4.08
2	2	Jeans	11191.69	211550	4.10
3	3	Backpack	11044.13	386285	4.16
4	4	Sweatshirt	10735.30	210051	4.16
5	5	Smartphone	10713.56	599483	4.15
6	6	Laptop	10686.72	338004	4.13
7	7	Travel Bag	10680.59	331011	4.15
8	8	Hoodie	10655.74	177853	4.27
9	9	Training Shoes	10361.42	296865	4.04
10	10	Headphones	10207.61	180266	4.13
11	11	T-shirt	10190.95	176916	4.05
12	12	Sneakers	10013.36	597227	4.04
13	13	Sunglasses	9967.97	329471	4.15
14	14	Jacket	9888.80	206222	4.04
15	15	Earbuds	9644.56	165733	4.08
16	16	Watch	9546.70	297931	4.05
17	17	Running Shoes	9469.65	308958	4.00

- **High-Engagement Engines:** *Smartphone* and *Sneakers* are massive transaction drivers, accumulating nearly 600,000 total reviews each, proving they command the highest customer purchase volumes on the platform.

- **Underappreciated Superstars:** *Hoodie* ranks 8th in popularity but holds the highest customer satisfaction score (4.27) on the chart, signaling a highly reliable product category that customers love.

- **The Bottom Performers:** *Running Shoes* sit at the bottom of the list with the lowest average rating (4.00), signaling potential quality or vendor sizing issues.

Action recommendation:

- **De-risk Core Volume Subcategories:** Ensure *Smartphone* and *Sneakers* subcategories have flawless user experience pipelines, as any drop in their search visibility or inventory immediately threatens the platform's core traffic.

- **Audit Low-Performing Lines:** Investigate individual products under the *Running Shoes* subcategory to identify which specific vendors are bringing down the average ratings.

7. Products with high price and low popularity

Identifying premium, expensive items that are failing to generate customer engagement or platform traffic, highlighting potential overpricing or lack of market demand.

	product character varying (100) 🔒	category character varying (50) 🔒	price numeric (10,2) 🔒	popularity_score numeric (10,2) 🔒
1	Dell Laptop	Electronics	79156.00	8811.00
2	Sony Headphones	Electronics	78684.00	7908.90
3	HP Pavilion	Electronics	78600.00	1472.20
4	OnePlus 11	Electronics	78341.00	6375.60
5	OnePlus 11	Electronics	78133.00	5655.60
6	iPhone 13	Electronics	77778.00	3609.60
7	Dell Laptop	Electronics	77753.00	3061.60
8	OnePlus 11	Electronics	77671.00	836.60
9	Boat Earbuds	Electronics	77454.00	9613.50
10	Dell Laptop	Electronics	76168.00	8697.60
11	OnePlus 11	Electronics	75889.00	5184.20
12	OnePlus 11	Electronics	75594.00	6220.50
13	iPhone 13	Electronics	74501.00	5456.50
14	HP Pavilion	Electronics	74443.00	9856.80
15	iPhone 13	Electronics	74316.00	7552.90
16	iPhone 13	Electronics	74255.00	792.00
17	Boat Earbuds	Electronics	73928.00	6485.60

- **Electronics Concentration:** The list is entirely populated by Electronics, which tie up significant capital.
- **Severe Outliers (Dead Stock Risk):** Specific variants like the *OnePlus 11* (77,671.00 with a popularity score of only 836.60) and *iPhone 13* (74,255.00 with a popularity score of 792.00) are practically invisible to customers despite their premium pricing.
- **Mismatched Value Perception:** Typically budget-friendly subcategories like *Boat Earbuds* are listed at extreme premium price points (over 73,000+), resulting in lowered popularity scores relative to standard inventory.

Action recommendation:

- **Implement Targeted Price Cuts:** Deploy flash discounts or trade-in offers specifically for the low-popularity/high-price variants (like the low-performing HP Pavilion and OnePlus 11 listings) to liquidate stagnant inventory.
- **Correct Listing Pricing Inconsistencies:** Audit listings where standard brand expectations conflict with high data pricing (e.g., Boat Earbuds at premium laptop rates) to fix potential manual seller pricing errors.

8. Products with high discount but low rating.

Identifying underperforming or low-quality products that sellers are aggressively discounting (55% to 70%) to liquidate stock, risking platform reputation and high return rates.

	product character varying (100)	category character varying (50)	discount_percentage integer	price numeric (10,2)	rating numeric (3,1)
1	Fossil Watch	Accessories	70	3720.00	3.2
2	Adidas Ultraboost	Footwear	69	3024.00	3.2
3	Fossil Watch	Accessories	68	4896.00	3.2
4	RayBan Sunglasses	Accessories	68	1672.00	3.2
5	Fossil Watch	Accessories	65	5714.00	3.2
6	Reebok Shoes	Footwear	65	7740.00	3.2
7	Samsung Galaxy S21	Electronics	64	45056.00	3.2
8	Nike T-shirt	Fashion	63	762.00	3.2
9	American Tourister Bag	Accessories	63	1017.00	3.2
10	Adidas Hoodie	Fashion	62	1511.00	3.2
11	iPhone 13	Electronics	61	73717.00	3.2
12	H&M Sweatshirt	Fashion	60	3640.00	3.2
13	American Tourister Bag	Accessories	59	2480.00	3.2
14	Boat Earbuds	Electronics	58	20100.00	3.2
15	Reebok Shoes	Footwear	58	3638.00	3.2
16	American Tourister Bag	Accessories	57	2449.00	3.2
17	Puma Sneakers	Footwear	55	5199.00	3.2

- **The "Discount Trap":** These products possess a flat 3.2 rating despite steep price cuts, indicating that a low price point is not enough to overcome underlying issues with product quality or fulfillment accuracy.
- **Capital Exposure Risk:** High-ticket electronics like the *iPhone 13* (73,717.00 at 61% off) and *Samsung Galaxy S21* (45,056.00 at 64% off) tied to

low consumer satisfaction represent high return rates and customer frustration risks.

Action recommendation:

- **Quality and Listing Audits:** Pause algorithmic promotion of these items and conduct catalog quality checks to understand if the poor ratings stem from damaged inventory, sizing mismatches, or counterfeit listings.
- **Transition to Clearance Badging:** Relabel these specific inventory lines from standard promotional campaigns to "Clearance/As-Is" status to set appropriate consumer expectations and protect the platform's brand equity.

9. Out-of-stock products with high popularity or high reviews.

Pinpointing the highest-demand products that are currently unavailable, which represents an immediate leakage of platform revenue and a frustrating experience for high-intent shoppers.

	product character varying (100) 🔒	stock_status character varying (20) 🔒	popularity_score numeric (10,2) 🔒	reviews_count integer 🔒
1	Wildcraft Backpack	Out of Stock	23947.20	4989
2	OnePlus 11	Out of Stock	23902.20	4878
3	Puma Jacket	Out of Stock	23605.00	4721
4	Sony Headphones	Out of Stock	23601.60	4917
5	OnePlus 11	Out of Stock	23582.40	4913
6	Sony Headphones	Out of Stock	23260.00	4652
7	iPhone 13	Out of Stock	22755.60	4644
8	RayBan Sunglasses	Out of Stock	22696.30	4829
9	iPhone 13	Out of Stock	22612.80	4711
10	American Tourister Bag	Out of Stock	22409.60	4768
11	iPhone 13	Out of Stock	22291.60	4846
12	American Tourister Bag	Out of Stock	22089.20	4508
13	Puma Jacket	Out of Stock	21956.90	4481
14	Wildcraft Backpack	Out of Stock	21734.40	4528
15	Adidas Ultraboost	Out of Stock	21631.50	4807
16	iPhone 13	Out of Stock	21474.30	4569
17	Fossil Watch	Out of Stock	21450.00	4290

- **Severe Revenue Bleed:** This list features multiple entries for core items like the *iPhone 13*, *OnePlus 11*, *Wildcraft Backpack*, and *Sony Headphones*, proving that systemic stockouts are occurring across entire high-performing product lines rather than just isolated SKUs.
- **High-Intent Traffic Drop-Off:** Every single product here combines extreme popularity scores (>21,000\$) with deep review history (>4,000\$), meaning organic search traffic is landing on dead ends.

Action recommendation:

- **Deploy "Notify Me" and Backorder Widgets:** Immediately activate an automated email/SMS waitlist on these specific product pages to capture and retain customer intent while inventory is restored.
- **Prioritize Tier-1 Replenishment Logs:** Establish a priority replenishment flag in the logistics system explicitly for these 17 high-traffic items to automate emergency reorders before stock completely zeroes out.

10. Monthly product additions and trend by category.

Tracking catalog expansion trends over time to understand seller upload behavior and identify which product categories are scaling the fastest.

	uploaded_month text	category character varying (50)	product_added bigint
1	2024-05	Accessories	25
2	2024-05	Footwear	18
3	2024-05	Electronics	17
4	2024-05	Fashion	13
5	2024-06	Footwear	26
6	2024-06	Accessories	25
7	2024-06	Fashion	22
8	2024-06	Electronics	20
9	2024-07	Footwear	24
10	2024-07	Accessories	23
11	2024-07	Electronics	22
12	2024-07	Fashion	14
13	2024-08	Footwear	29
14	2024-08	Accessories	24
15	2024-08	Electronics	23
16	2024-08	Fashion	21
17	2024-09	Fashion	23

- **Footwear Core Expansion:** *Footwear* demonstrates consistent, aggressive growth, peaking with 29 new additions in August 2024, signalling strong vendor supply lines in this segment.
- **Steady Accessories Baseline:** *Accessories* maintains a stable and predictable upload rhythm, averaging around 23 to 25 items monthly from May to August.
- **Laggard Fashion Slumps:** *Fashion* updates are historically volatile, dipping significantly in July (14 additions) before seeing a recovery heading into September.

Action recommendation:

- **Align Marketing Calendars with Upload Peaks:** Sync high-budget promotional campaigns with known inventory influxes (e.g., scale up Footwear marketing in late August to immediately push newly listed stock).
- **Incentivize Off-Season Uploads:** Offer seller fee discounts or visibility boosts to Fashion vendors during historical low-upload months (like July) to stabilize catalog growth variations throughout the year.

11. Products added recently but already performing well

Identifying newly introduced catalog products that have quickly secured exceptional customer satisfaction and popularity, pointing to immediate market fit and viral potential.

	product character varying (100) 🔒	rating numeric (3,1) 🔒	popular numeric (10,2) 🔒
1	Adidas Ultraboost	5.0	24015.00
2	Levi's Jeans	5.0	23965.00
3	Adidas Hoodie	5.0	17155.00
4	Reebok Shoes	5.0	13465.00
5	Wildcraft Backpack	5.0	12705.00
6	RayBan Sunglasses	5.0	12445.00
7	HP Pavillion	5.0	12330.00
8	Boat Earbuds	5.0	1510.00
9	H&M Sweatshirt	4.9	21530.60
10	Levi's Jeans	4.9	14964.60
11	iPhone 13	4.9	13259.40
12	Dell Laptop	4.9	10966.20
13	Adidas Ultraboost	4.9	10579.10
14	Reebok Shoes	4.9	9829.40
15	Nike Air Max	4.9	6007.40
16	American Tourister Bag	4.9	5943.70
17	American Tourister Bag	4.9	1327.90

- **Instant Market Fit:** Products like *Adidas Ultraboost* and *Levi's Jeans* have achieved a perfect 5.0 rating alongside top-tier popularity scores shortly after launch, demonstrating massive immediate customer traction.
- **High-Velocity Contenders:** The presence of high-ticket items (*HP Pavilion*, *iPhone 13*) with a 4.9–5.0 rating trend indicates strong introductory momentum for premium hardware additions.

Action recommendation:

- **Fast-Track Inventory Pipeline:** Immediately establish dedicated reorder cycles for these fast-moving new listings to prevent stock depletion before they hit maturity.
- **Amplify Through Collection Placement:** Bundle these trending items into curated "New Arrivals & Fan Favorites" marketing carousels on the platform home page to maintain velocity.

12. Products with the biggest gap between price and discounted price

Isolating items undergoing the most extreme price cuts (absolute margin drops) to track margin erosion and identify heavy promo-dependent listings.

	product character varying (100) 🔒	price numeric (10,2) 🔒	discounted_price numeric (10,2) 🔒	gap numeric 🔒
1	HP Pavilion	79814.00	23944.20	55869.80
2	Boat Earbuds	75153.00	23297.43	51855.57
3	Sony Headphones	72196.00	22380.76	49815.24
4	Dell Laptop	70947.00	21993.57	48953.43
5	Samsung Galaxy S21	74168.00	25958.80	48209.20
6	OnePlus 11	78691.00	30689.49	48001.51
7	iPhone 13	79407.00	31762.80	47644.20
8	Dell Laptop	67724.00	20317.20	47406.80
9	HP Pavilion	68281.00	21167.11	47113.89
10	iPhone 13	73614.00	26501.04	47112.96

- **Massive Margin Sacrifices:** High-end electronics are taking absolute price cuts of over 47,000 Rs to ₹55,000 Rs, signalling desperate inventory liquidation or hyper-aggressive customer acquisition loss-leaders.

- **Pricing Identity Inconsistencies:** Similar to Query 7, mid-tier items like *Boat Earbuds* are listed with inflated original baseline prices (~75,153.00 Rs) just to display a massive absolute discount gap.

Action recommendation:

- **Re-evaluate Margin Health:** Audit the gross profit margins for the *HP Pavilion* and *Dell Laptop* lines to ensure these huge absolute gaps aren't causing net financial losses.
- **Create a "Mega Deals" Section:** Group these highest-gap items into a dedicated high-impact deal banner on the homepage to maximize high-conversion click-through rates.

13. Brands with the highest share of out-of-stock items

Isolating vendor supply chain bottlenecks by identifying which brands suffer from the highest proportional stockout rates, leading to lost customer momentum.

	brand character varying (50) 🔒	count bigint 🔒	out_of_stock_count bigint 🔒	out_of_stock_percentage numeric 🔒
1	Zara	78	33	42.31
2	Adidas	203	83	40.89
3	OnePlus	77	31	40.26
4	Samsung	74	27	36.49
5	Wildcraft	146	53	36.30
6	Apple	83	30	36.14
7	Boat	70	25	35.71
8	Reebok	116	40	34.48
9	RayBan	137	43	31.39
10	HP	64	20	31.25
11	Fossil	128	40	31.25
12	American Tourister	129	40	31.01
13	H&M	82	25	30.49
14	Nike	191	57	29.84
15	Puma	206	59	28.64
16	Levi's	77	22	28.57
17	Sony	73	19	26.03

- **Critical Inventory Bleed:** Fast-fashion giant *Zara* and athletic apparel leader *Adidas* are severely undersupplying the platform, with over **40% of their total catalog completely unavailable** for purchase.

- **Tech Logistics Snags:** Premium electronic manufacturers (*OnePlus*, *Samsung*, and *Apple*) are locking up customer wallets, with roughly 1 in every 3 products sitting on out-of-stock pages.

Action recommendation:

- **Enforce Out-of-Stock Penalties/Caps:** Introduce platform penalties or limit promotional visibility for vendors like *Zara* and *Adidas* if their total stockout rate remains above a 30% operational threshold.

- **Automate Reorder Lead Times:** Sync with *OnePlus*, *Samsung*, and *Apple* distribution channels to dynamically adjust procurement safety margins, accounting for longer tech-supply turn around periods.

14. Categories with the weakest average rating

Diagnosing macro-level consumer dissatisfaction by identifying which broad product categories are underperforming in quality, sizing accuracy, or performance metrics.

	category character varying (50) 🔒	weakest_rating numeric 🔒	toital_product bigint 🔒
1	Footwear	4.03	490
2	Fashion	4.12	463
3	Accessories	4.13	540
4	Electronics	4.13	507

- **Footwear Core Vulnerability:** *Footwear* exhibits the most pronounced quality slump with an average rating of 4.03 across a substantial volume of 490 products, signalling systemic platform-wide issues such as structural sizing mismatches, poor material quality, or misleading product imagery.

- **Narrow Quality Ceiling:** Interestingly, all major categories are clustered tightly within a narrow band of 4.03 to 4.13, which means no single sector is distinctively running away as a "gold standard" for consumer satisfaction.

Action recommendation:

- **Deploy Sizing Intelligence and Fit Charts:** For the bottom-ranking *Footwear* and *Fashion* sectors, integrate dynamic customer feedback widgets (e.g., "Runs True to Size", "Runs Small") on product pages to actively reduce return rates driven by fit expectations.
- **Filter Out Underperforming Low-Tier Vendors:** Conduct an automated purge or issue performance warnings to individual sellers whose consistent sub-3.5 ratings are dragging down the macro averages of the *Footwear* and *Fashion* categories.

15. Best and worst products in each brand.

Evaluating intra-brand quality variance by juxtaposing a manufacturer's highest-rated product against their lowest-performing offering to expose brand inconsistency or manufacturing defects.

	brand character varying (50) 🔒	product character varying (100) 🔒	best_rating numeric (3,1) 🔒	worst_product character varying (100) 🔒	worst_rating numeric (3,1) 🔒
1	Adidas	Adidas Ultraboost	5.0	Adidas Hoodie	3.2
2	American Tourister	American Tourister Bag	5.0	American Tourister Bag	3.2
3	Apple	iPhone 13	5.0	iPhone 13	3.2
4	Boat	Boat Earbuds	5.0	Boat Earbuds	3.2
5	Dell	Dell Laptop	5.0	Dell Laptop	3.3
6	Fossil	Fossil Watch	5.0	Fossil Watch	3.2
7	H&M	H&M Sweatshirt	5.0	H&M Sweatshirt	3.2
8	HP	HP Pavilion	5.0	HP Pavilion	3.2
9	Levi's	Levi's Jeans	5.0	Levi's Jeans	3.3
10	Nike	Nike Air Max	5.0	Nike Air Max	3.2
11	OnePlus	OnePlus 11	4.9	OnePlus 11	3.3
12	Puma	Puma Jacket	5.0	Puma Sneakers	3.2
13	RayBan	RayBan Sunglasses	5.0	RayBan Sunglasses	3.2
14	Reebok	Reebok Shoes	5.0	Reebok Shoes	3.2
15	Samsung	Samsung Galaxy S21	5.0	Samsung Galaxy S21	3.2
16	Sony	Sony Headphones	5.0	Sony Headphones	3.2
17	Wildcraft	Wildcraft Backpack	5.0	Wildcraft Backpack	3.2
18	Zara	Zara Shirt	5.0	Zara Shirt	3.2

- **Severe SKU-Level Quality Gaps:** Brands like *Apple (iPhone 13)*, *Samsung (Galaxy S21)*, and *Sony (Sony Headphones)* show the exact same product name taking both a perfect 5.0 and a low 3.2/3.3 rating. This reveals that consumer satisfaction isn't breaking down at the brand level, but rather at the **individual seller/variant level** (e.g., bad batches, renewed/refurbished listings mixed with new ones, or fulfillment errors).
- **Uniform Satisfaction Ceilings & Floors:** Across almost every single brand listed, the best products confidently reach a ceiling of 5.0 (or 4.9 for OnePlus), while the worst bottom out consistently at a tight floor of 3.2 to 3.3.

Action recommendation:

- **Isolate Variant and Seller IDs:** Modify the underlying query to drill down past generic product names into specific SKU, Seller, or Condition codes (New vs Refurbished) to find exactly who is selling the 3.2-rated iPhone 13s or Galaxy S21s.
- **Establish Brand Consistency Alerts:** Work with major partners like Nike, Adidas, and Apple to investigate their lower-tier variants (like the Adidas Hoodie) or problematic seller channels to protect their overall brand equity on your marketplace.

Conclusion

In conclusion, this analysis highlights that the e-commerce business's main challenge is not only identifying strong products, but also ensuring that high-demand items remain available, appropriately priced, and supported by consistent inventory planning. The findings show that several brands and categories generate strong customer interest, but stock shortages and pricing imbalances are limiting their full business potential. At the same time, some high-priced products are not performing well, which suggests the need for closer review of pricing strategy, product quality, and promotional effectiveness. Overall, the most effective business response is to prioritize restocking for high-performing products, refine pricing and discount decisions for weak listings, and focus marketing and operational support on the brands and categories that consistently deliver customer engagement and satisfaction.